Heming Qin

Software Engineer | Full-Stack & Backend Development | IT Systems
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SUMMARY

Software engineer experienced in Android applications, IT systems, full-stack web tools, backend APIs, and AI-assisted search systems. Developed carrier integrations, real-time services, and data pipelines with Java, Kotlin, Python, React, and Flask. Experienced in debugging, automated testing, and cloud deployment with AWS, Docker, and CI/CD.

EDUCATION

UNIVERSITY OF VICTORIA
Master of Applied Science in Electrical and Computer Engineering Jan 2026

SOUTHEAST UNIVERSITY
Master of Engineering, Optical Engineering Jun 2020

XI'AN UNIVERSITY OF POSTS AND TELECOMMUNICATIONS
Bachelor of Engineering, Optoelectronic Information Science and Technology Jul 2017

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, SQL, Bash/Shell Scripting, HTML5, CSS3

Frontend & Full-Stack: React, Web Application Development, Responsive Web Design, API Integration

Backend & Architecture: Flask, FastAPI, RESTful APIs, WebSocket, Distributed Systems, Event-Driven Architecture, Asynchronous Processing

Data & AI: PostgreSQL, MongoDB, Redis, SQL Query Optimization, Vector Embeddings, Retrieval-Augmented Generation (RAG), Semantic Search, Data Pipelines

Cloud & DevOps: AWS EC2, Docker, Nginx, Linux, GitHub Actions, CI/CD, Git

Software Engineering & IT: System Design, Android SDK, JUnit, Unit Testing, Integration Testing, Authentication & Authorization, Access Control, Troubleshooting, Agile/Jira

Soft Skills: Effective Communication, Team Player, Team Leadership, Mentoring, Problem-Solving, Resilient

PROFESSIONAL EXPERIENCE

University of Victoria | Research Software Developer
Jul 2024 – Jan 2026

- Applied software development to scientific research, helping researchers analyze simulation results and work with technical data.
- Built a research knowledge and simulation platform that brought simulation results, technical documents, and research records into one searchable application.
- Developed automated data ingestion and metadata pipelines for simulation outputs and research documents, reducing manual organization work by approximately 40%.
- Implemented backend APIs for semantic search and Retrieval-Augmented Generation (RAG) using PostgreSQL and vector embeddings to retrieve historical simulation data and technical records.
- Maintained Docker-based data processing and deployment workflows using AWS EC2, GitHub Actions, and YAML.
- Worked with researchers to understand how they used technical data and develop search features for past simulation results.

OPPO Mobile Telecommunications Corp | Software Developer
Jan 2021 – Aug 2023

- Developed and tested Android security features to meet carrier requirements across devices and markets.
- Served as technical lead for five engineers developing LockAssistant, an Android application used by millions of people in more than 20 countries.
- Developed Java and Kotlin Android framework components and asynchronous services for SIM-based authorization, carrier provisioning, device security, and access control.

PROFESSIONAL EXPERIENCE (CONTINUED)

OPPO Mobile Telecommunications Corp | Software Developer Jan 2021 – Aug 2023

- Integrated third-party SDKs and built cross-device features for notifications, file sharing, and communication between smartphones, tablets, and smart TVs.
- Debugged Android platform and carrier-certification failures, tracing lock-screen, UI, and state-transition issues to their root causes and verifying fixes with QA and carrier partners.
- Wrote JUnit unit and regression tests and used Gradle and shell scripts to support build validation, automated checks, and production releases.
- Maintained carrier-specific Android code through major OS upgrades, identifying compatibility issues and validating changes across devices and software configurations.

OnePlus Software Technology Ltd | Software Developer Jul 2020 – Jan 2021

- Improved Android calling, messaging, and accessibility features for customers on different networks.
- Developed and maintained Android communication features in Java and Kotlin, including Real-Time Text (RTT) and Rich Communication Services (RCS) across OS versions.
- Built speech-to-text and text-to-speech accessibility features to make Android communication apps easier to use for people with hearing or visual impairments.
- Integrated Android Telecom Framework and Binder-based inter-process communication (IPC) across Contacts, Messaging, carrier services, and modem interfaces.
- Ran carrier certification and integration tests, debugging RTT/RCS activation, communication state changes, and session synchronization issues.
- Improved Android app performance and stability by simplifying interface layouts and fixing lifecycle and permission-handling issues.

ADDITIONAL EXPERIENCE

TD Bank | Customer Experience Associate Sep 2026 – Present

- Explained digital banking services in everyday language and helped customers work through account problems.
- Troubleshoot mobile and online banking issues, helping customers navigate digital platforms and resolve account-related problems.
- Handle sensitive financial information while maintaining data accuracy, privacy, and compliance with banking procedures.
- Explain digital banking steps to customers with different levels of technical experience and work with colleagues to resolve issues.

Mercor | Software Developer – AI Model Evaluation Jun 2026 – Sep 2026

- Checked AI-generated answers for errors and documented findings to help improve model quality.
- Reviewed large language model (LLM) outputs for factual accuracy, reasoning errors, and consistency; recorded findings through structured technical evaluations.

VOLUNTEER EXPERIENCE

Oak Bay Volunteer Services (OBVS) | Volunteer Software Developer Mar 2025 – Jun 2025

- Developed web features based on volunteer and staff needs, making day-to-day coordination easier.
- Built a full-stack web application with React, JavaScript, Python, Flask, PostgreSQL, and REST APIs for volunteer coordination, donation management, and internal workflows.
- Translated volunteer and staff workflows into role-based web features, connecting React interfaces with Flask backend APIs and database records.
- Replaced polling with WebSocket and Redis Pub/Sub to deliver real-time notifications and synchronization for 70+ concurrent users.
- Reduced backend API response times by approximately 30% through PostgreSQL indexing, SQL query tuning, and stateless API design.
- Deployed containerized services using Docker, Nginx, AWS EC2, and GitHub Actions CI/CD pipelines.